FutureXcel Internship Program

Title:

Complete Exploratory Data Analysis (EDA) Using Superstore Dataset

Role:

Data Science Intern

Organization:

FutureXcel

Week: 3

Complete Exploratory Data Analysis (EDA)

Prepared by:

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Tools & Technologies:

Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

Submission Date:

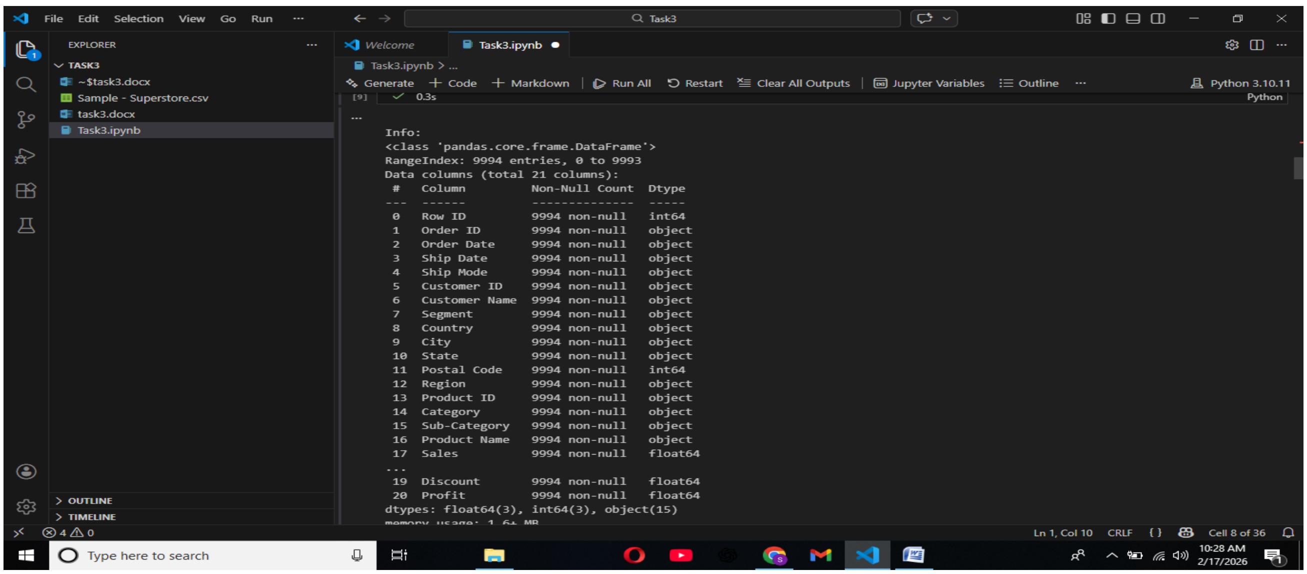
17 February 2026

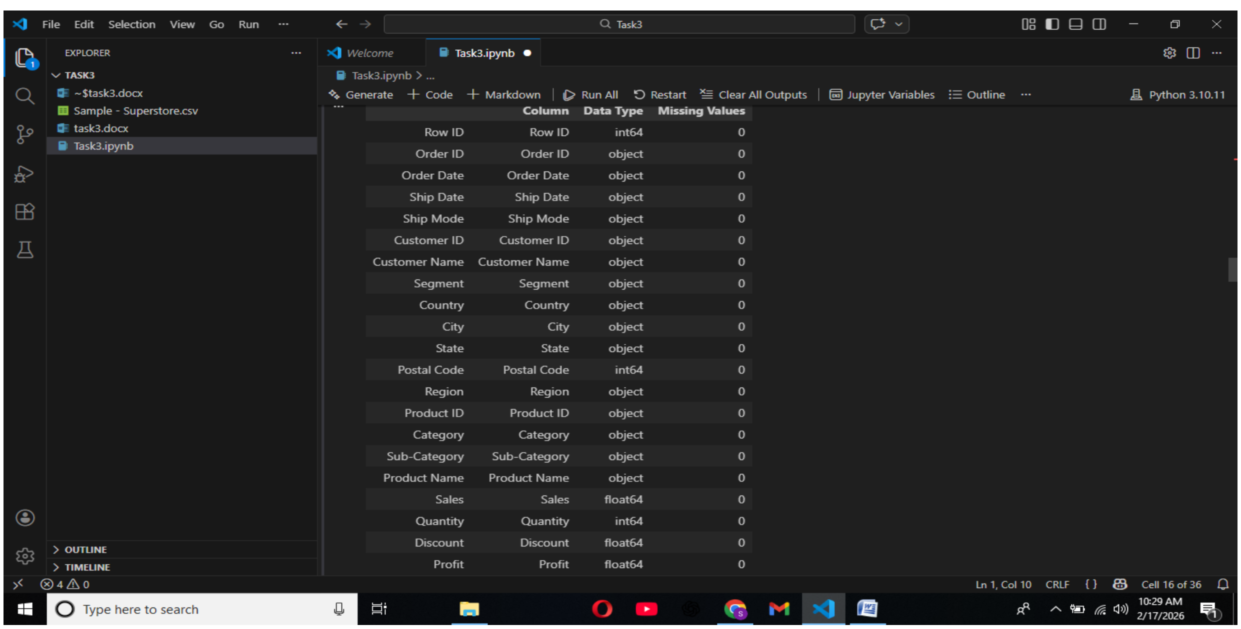
1. Dataset Overview

The dataset contains retail sales transactions of a superstore.
It includes information about orders, customers, regions, products, sales, profit, and discounts.

- Total Rows: 9994
- Total Columns: 21
- Missing Values: 0
- Numeric Columns: Sales, Profit, Discount, Quantity

The dataset is clean and ready for analysis.





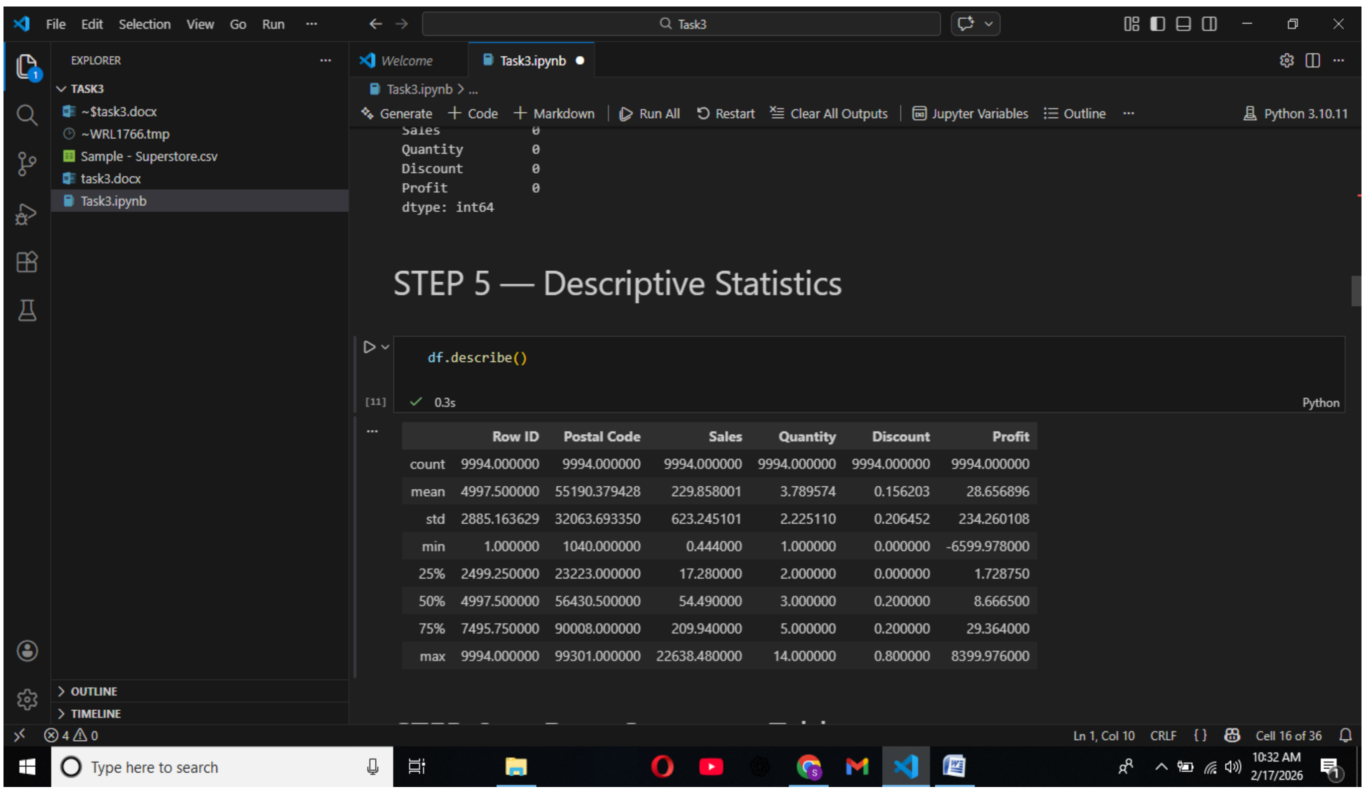
	Column	Data Type	Missing Values
Row ID	Row ID	int64	0
Order ID	Order ID	object	0
Order Date	Order Date	object	0
Ship Date	Ship Date	object	0
Ship Mode	Ship Mode	object	0
Customer ID	Customer ID	object	0
Customer Name	Customer Name	object	0
Segment	Segment	object	0
Country	Country	object	0
City	City	object	0
State	State	object	0
Postal Code	Postal Code	int64	0
Region	Region	object	0
Product ID	Product ID	object	0
Category	Category	object	0
Sub-Category	Sub-Category	object	0
Product Name	Product Name	object	0
Sales	Sales	float64	0
Quantity	Quantity	int64	0
Discount	Discount	float64	0
Profit	Profit	float64	0

2. Descriptive Statistics

Descriptive statistics help us understand the numerical distribution of the dataset.

- Average Sales: 229
- Average Profit: 28
- Maximum Sales: 22638
- Minimum Profit: -6599

Sales and profit contain extreme values which indicate the presence of outliers.



	Row ID	Postal Code	Sales	Quantity	Discount	Profit
count	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000
mean	4997.500000	55190.379428	229.858001	3.789574	0.156203	28.656896
std	2885.163629	32063.693350	623.245101	2.225110	0.206452	234.260108
min	1.000000	1040.000000	0.444000	1.000000	0.000000	-6599.978000
25%	2499.250000	23223.000000	17.280000	2.000000	0.000000	1.728750
50%	4997.500000	56430.500000	54.490000	3.000000	0.200000	8.666500
75%	7495.750000	90008.000000	209.940000	5.000000	0.200000	29.364000
max	9994.000000	99301.000000	22638.480000	14.000000	0.800000	8399.976000

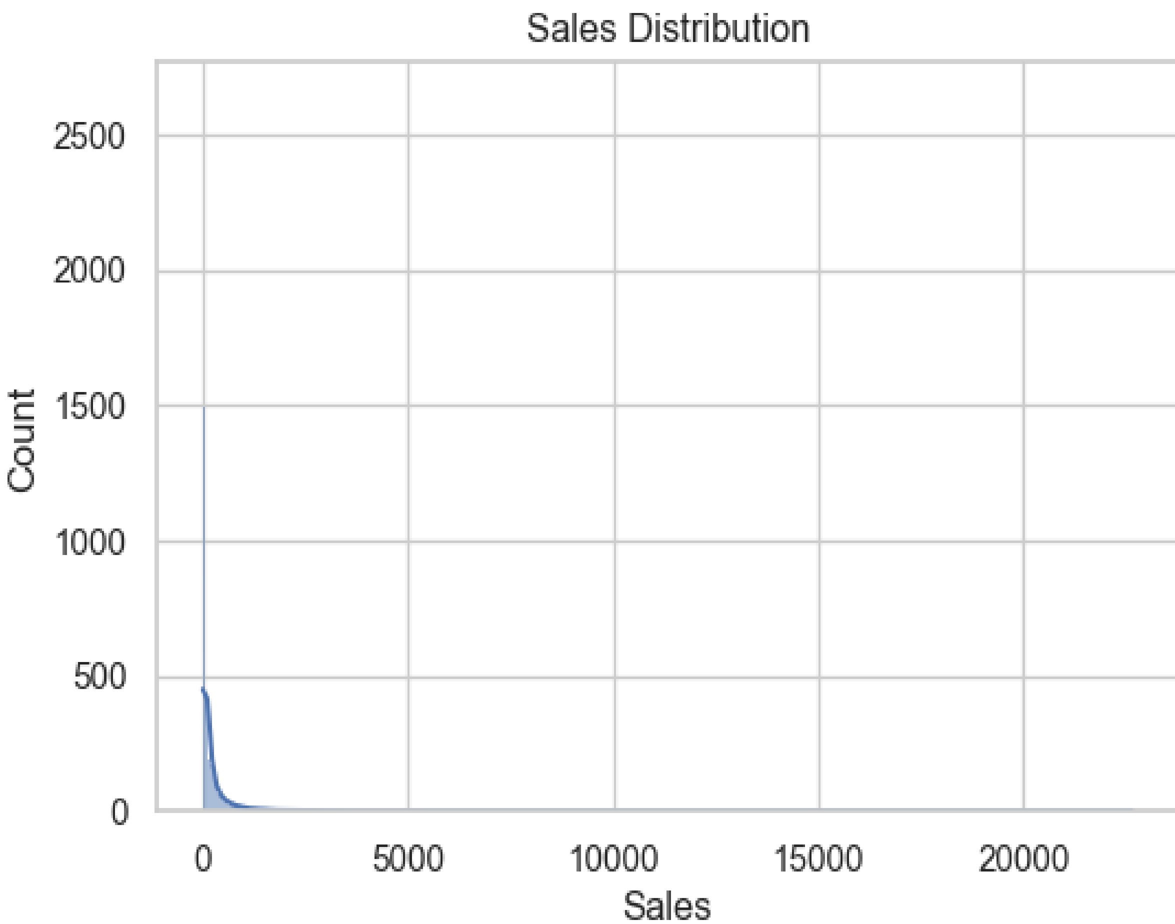
DISTRIBUTION ANALYSIS

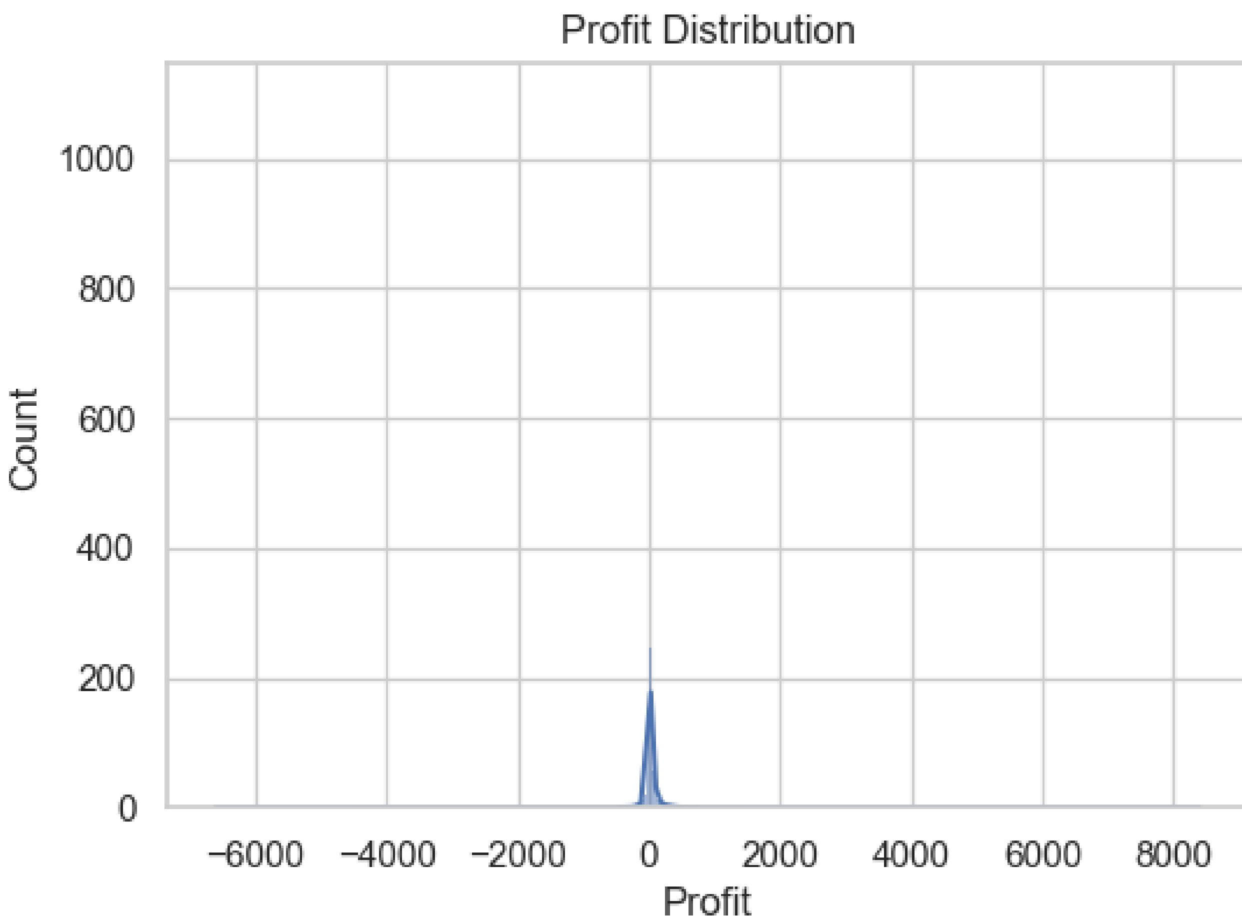
3. Distribution Analysis

The distribution of Sales and Profit is highly right-skewed.

- Sales Skewness: 12.97
- Sales Kurtosis: 305
- Profit Skewness: 7.56
- Profit Kurtosis: 397

High skewness indicates that most values are small while a few are extremely large.
High kurtosis indicates the presence of extreme outliers.






```python
print("Sales Skew:", df['Sales'].skew())
print("Sales Kurtosis:", df['Sales'].kurt())

print("Profit Skew:", df['Profit'].skew())
print("Profit Kurtosis:", df['Profit'].kurt())
```

[21] ✓ 0.0s Python

```
Sales Skew: 12.97275234181623
Sales Kurtosis: 305.311753246823
Profit Skew: 7.561431562468343
Profit Kurtosis: 397.1885145524141
```

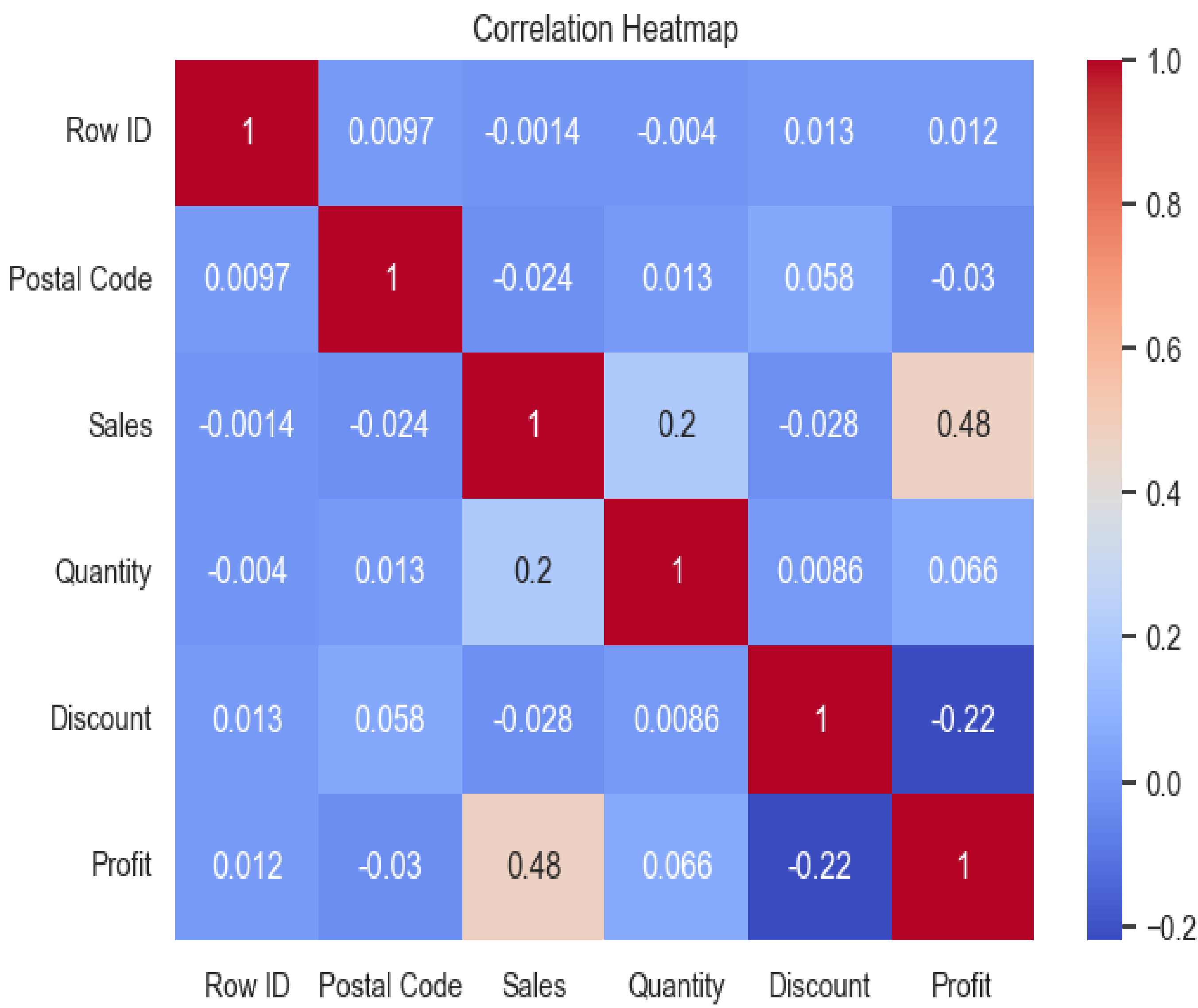
4. Correlation Analysis

The correlation heatmap shows relationships between numerical variables.

Key findings:

- Sales and Profit have a moderate positive correlation.
- Discount and Profit have a negative correlation.

This indicates that higher discounts often reduce profit.

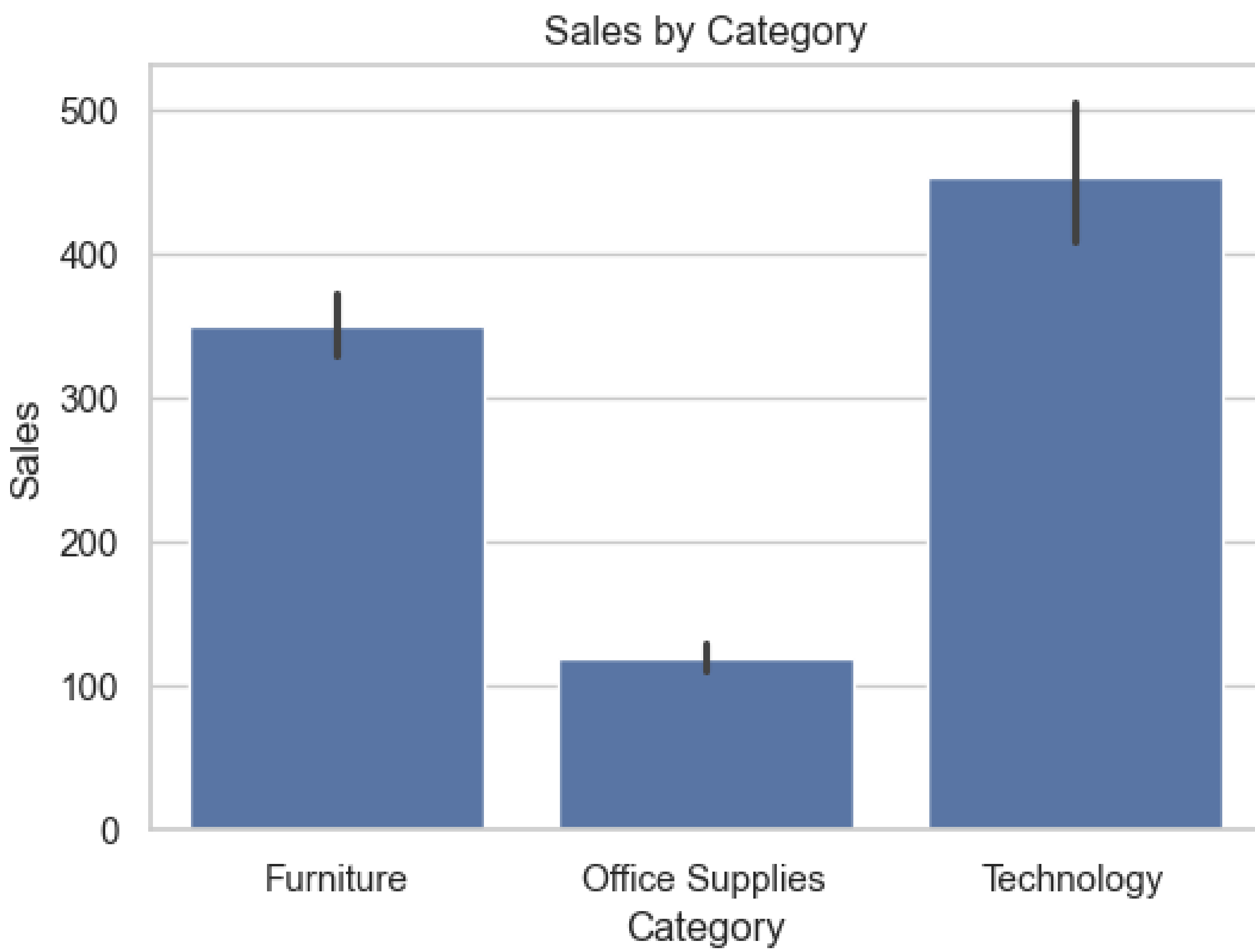


CATEGORY & REGION ANALYSIS

5. Category Analysis

Technology category generates the highest sales.
Office Supplies category generates the lowest sales.

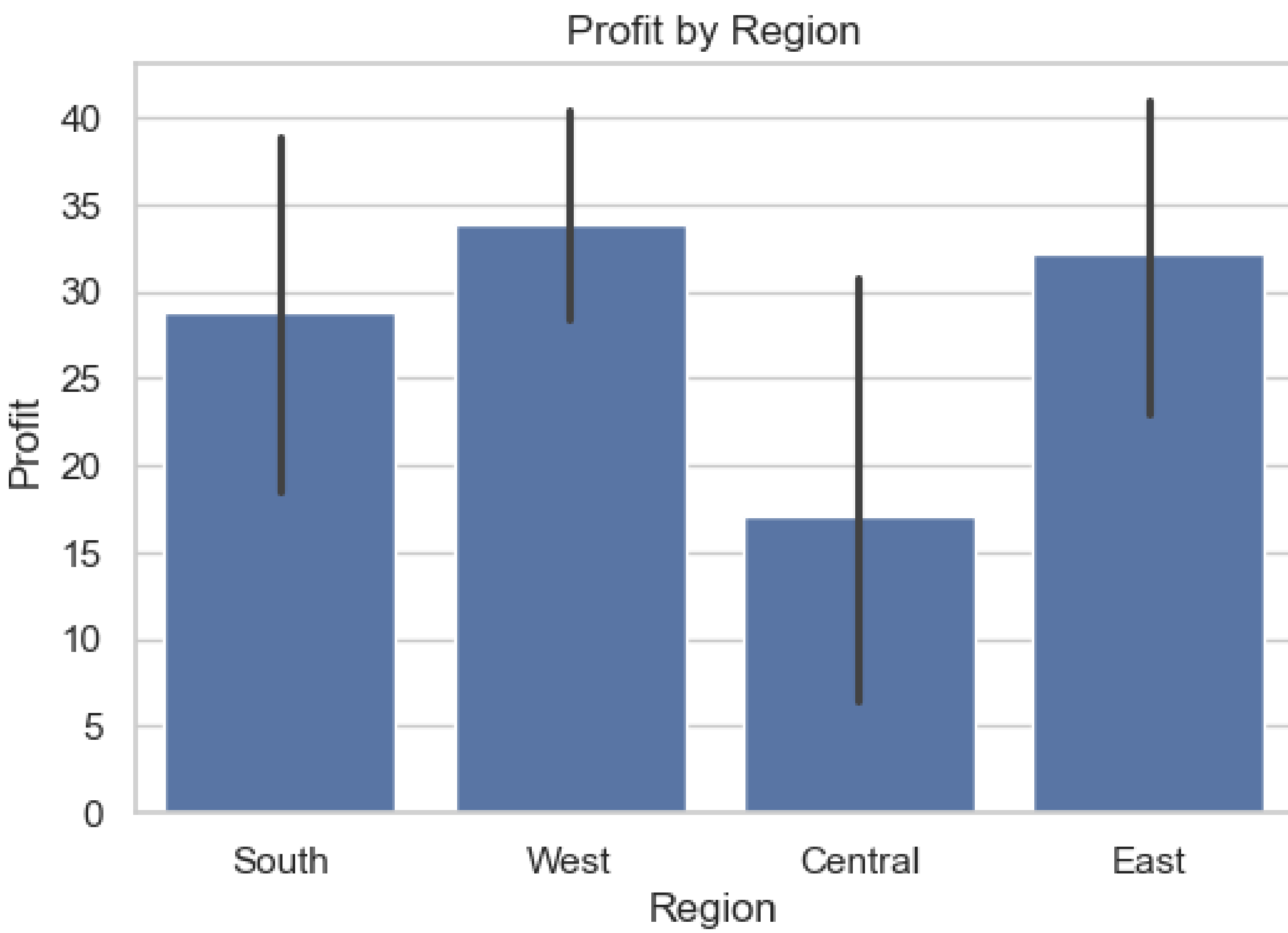
This suggests that technology products are more profitable for the business.



6. Region Analysis

The Central region shows the lowest profit.
The West region shows the highest profit.

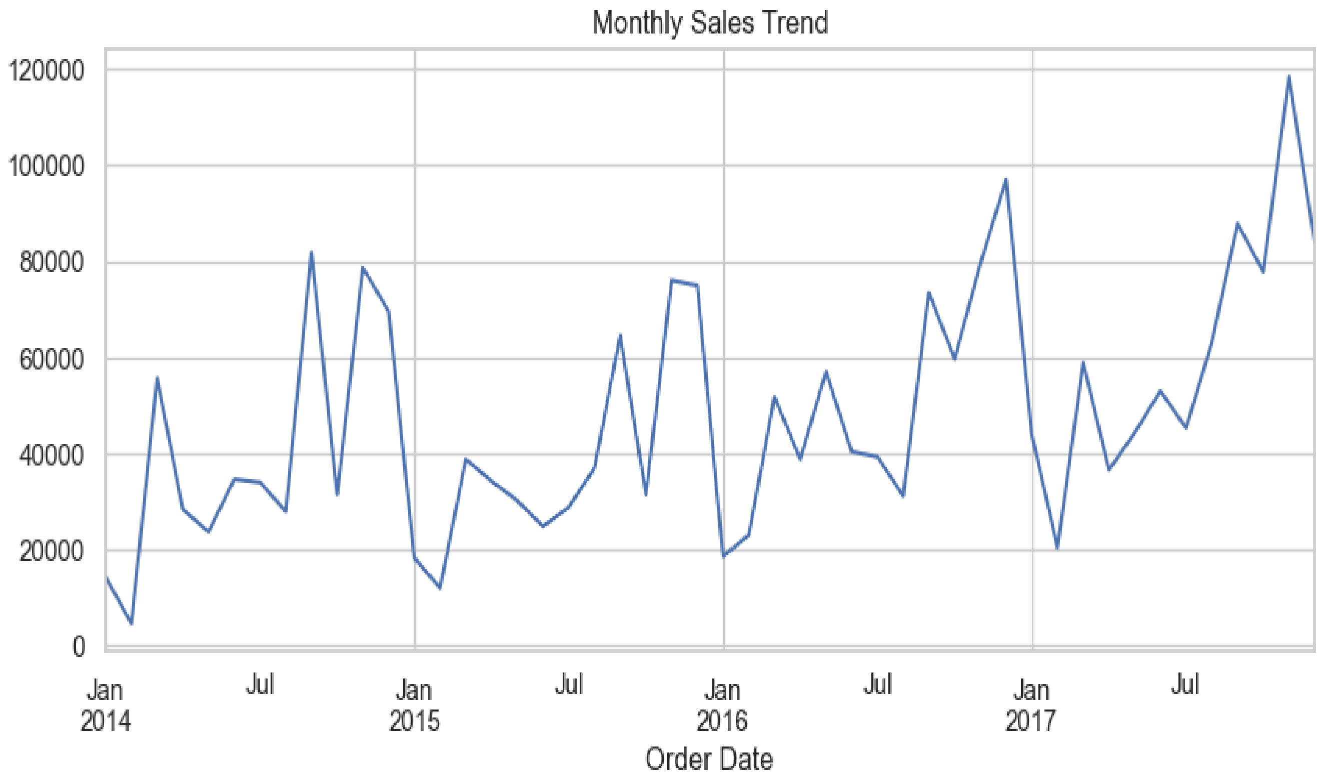
This suggests that business strategy in the Central region may need improvement.



7. Time Series Analysis

Monthly sales trend shows fluctuations across different months.
Some months have significantly higher sales compared to others.

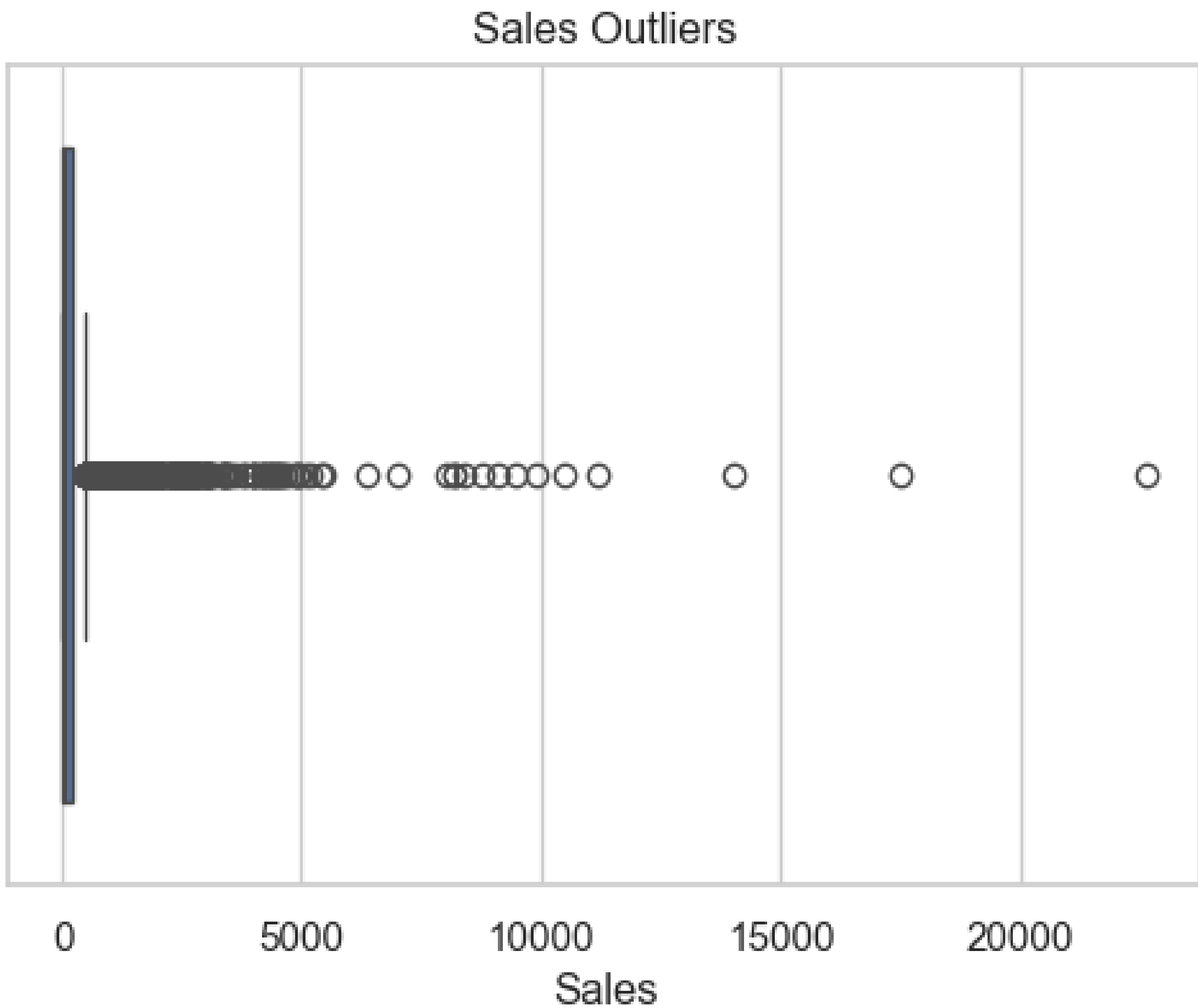
This helps identify seasonal trends and peak sales periods.



OUTLIERS & INSIGHTS

8. Outlier Analysis

Boxplot analysis shows extreme outliers in sales and profit.
These outliers represent very large transactions or potential anomalies.



9. Key Insights

1. Sales distribution is highly right-skewed with a few very large orders.
2. Profit distribution contains negative values indicating loss-making orders.
3. Sales and Profit show a moderate positive correlation.
4. Discount has a negative correlation with Profit.
5. Technology category generates the highest sales.
6. Office Supplies category generates the lowest sales.
7. Central region has the lowest profit.
8. The dataset contains extreme outliers in sales and profit.
9. Most orders have small quantity values.
10. The dataset is clean with no missing values.

10. Conclusion

The exploratory data analysis reveals key patterns in sales, profit, and discount behavior.
The business should focus on reducing excessive discounts and improving performance in low-profit regions.
Technology products contribute most to revenue and should be prioritized.